Goal and experimental design

Goal
Build a machine learning model to predict a player’s NBA Draft outcome group from pre-draft physical and performance data.

Prediction target
Multiclass classification, not exact pick prediction:
Top 5 • picks 6–14 • picks 15–30 • second round • undrafted

Models
Multinomial Logistic Regression
Naive Bayes
Decision Trees (ensemble methods)

Evaluation plan
Historical data: 2000–2025 draft players with known outcomes
2026 class: prediction pool
Main metric: macro F1-score
Additional checks: balanced accuracy, confusion matrix

Reasoning
Exact pick number is noisy and highly context-dependent. Broader draft outcome groups are more stable and easier to interpret.

Data

Dataset scope
Player-level dataset for NBA Draft classes 2000–2026
Size: ~2,350 players with 49 attributes
2026 players are included without final draft labels

Draft and combine data
NBA Draft History: draft year, overall pick, draft team, pre-draft organization
NBA Draft Combine (official pre-draft testing event for physical and athletic measurements): height, weight, wingspan, reach, vertical jump, agility, sprint, bench press

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College basketball performance data (NCAA)
Torvik: NCAA box-score and advanced statistics, mainly 2008–2026
Sports-Reference: NCAA statistics for early seasons 2000–2007 and early undrafted combine players

Additional planned enrichment:
NBA API: birth date → draft age

Known limitations for later preprocessing
Missing values are expected: some players skipped the combine, and international, high-school or club players do not have NCAA statistics. Cleaning, feature selection and imputation will be handled later.